
On the Limits of LLM Adaptability: Impact of Model-Internalized Priors on Annotation Task Performance

Etienne Casanova^{* 1} Rafal Kocielnik^{* 1} R. Michael Alvarez¹

Abstract

Large Language Models (LLMs) are increasingly used for zero-shot annotation and LLM-as-a-judge tasks, yet their reliability hinges on how model-internalized priors interact with user-provided instructions. We investigate three dimensions of this interaction: (1) how an LLM’s familiarity with data and task definitions relates to performance, (2) whether additional information in prompts can correct zero-shot errors (“decision stickiness”), and (3) model susceptibility to misaligned task definitions. We introduce Definition-Specific Familiarity (DSF), which measures alignment between a model’s elicited concept and the target definition. Across nine LLMs and five diverse toxicity datasets (spanning social media, gaming, news, and forums), DSF predicts model annotation performance after controlling for dataset identity (partial $r = +0.41$), and this association remains positive across all six prompting conditions tested. In contrast, three common text-memorization metrics show no positive association. We show that prompting has limited corrective power: only 34.8% of zero-shot errors are corrected by additional instructions or examples, with high-confidence errors especially persistent. Misaligned definitions systematically shift predictions without reducing reported confidence, making confidence unreliable for detecting definition-policy mismatch. Together, these findings establish definition alignment as a practical model-selection criterion and show that better prompting alone cannot substitute for validating model-policy fit.

^{*}Equal contribution. ¹California Institute of Technology, Pasadena, CA, USA. Correspondence to: Etienne Casanova <ecasanov@caltech.edu>, Rafal Kocielnik <rafalko@caltech.edu>.

1. Introduction

Large Language Models (LLMs) are increasingly deployed for zero-shot annotation (Gilardi et al., 2023) and rubric-based evaluation (LLM-as-a-judge) tasks (Zheng et al., 2023; Liu et al., 2023). These use cases are attractive: users provide a task definition via prompt, and the model annotates at scale without labeled training data (Gilardi et al., 2023; Brown et al., 2020). The implicit assumption is that the user’s definition governs the model’s behavior.

But LLMs are not blank slates. Through pre-training on web-scale corpora (Brown et al., 2020), followed by instruction tuning and reinforcement learning from human feedback (RLHF), models develop implicit understandings of common concepts and tasks. When a user asks an LLM to detect “toxic” content, the model has already been shaped by millions of documents and human-rated examples that discuss, define, and exemplify toxicity, and this exposure shapes an internal concept of the task. Critically, this internalized concept may not align with the user’s intended definition: toxicity is operationalized differently across applications, as identity attacks (Borkan et al., 2019), disruptive behavior in online gaming (Kocielnik et al., 2024; 2025b), or as speech targeting protected groups (ElSherief et al., 2018). An LLM anchored to a particular understanding may not simultaneously match all of these definitions, and its internalized priors may constrain how much user instructions can steer its behavior and alignment with the user’s intended interpretation (Min et al., 2022). This disconnect between a model’s internalized concepts and user-specified definitions represents a fundamental alignment challenge.

When the user’s task definition conflicts with the model’s internalized concept, which one governs behavior? Prior work has shown that LLMs exhibit biases reflecting the demographics and viewpoints in their training data — both at the pre-training stage and after instruction tuning or RLHF (Nadeem et al., 2021; Nangia et al., 2020; Parrish et al., 2022; Santurkar et al., 2023). Prompting and reasoning techniques can inadvertently amplify such latent biases (Shaikh et al., 2023). Less is known about the degree to which users can override a model’s internalized annotation task understanding through prompting, or how to measure the alignment between a model’s internal concept and a task

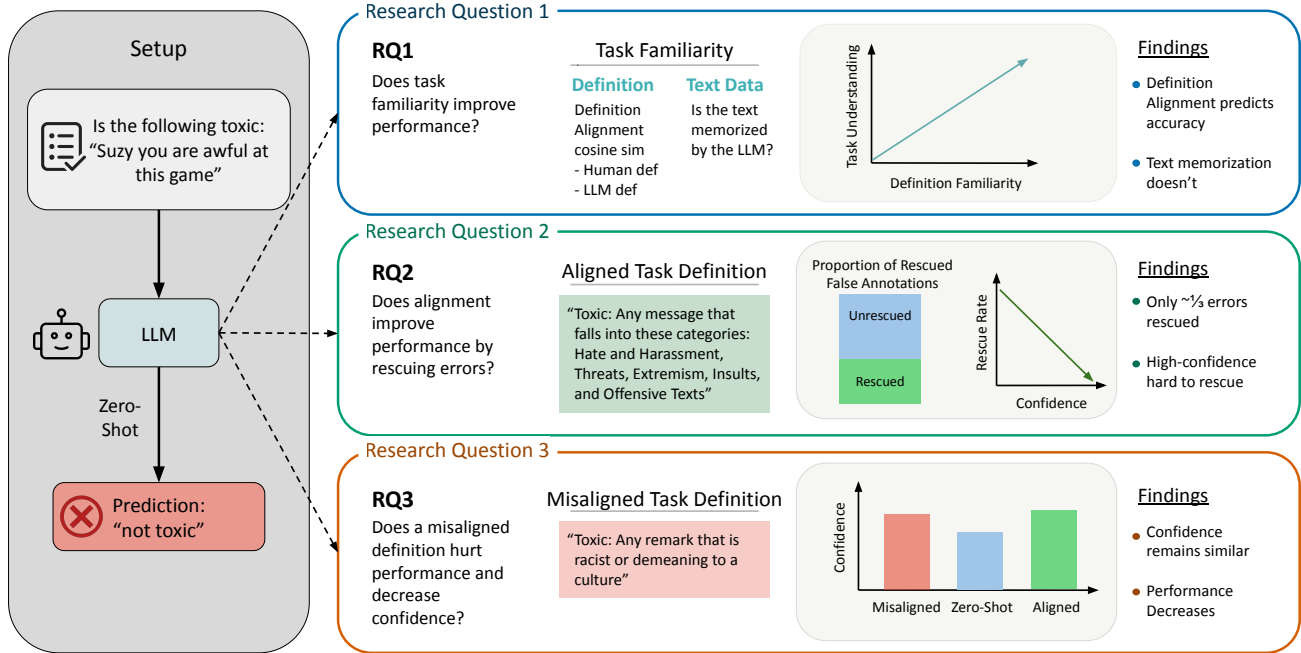


Figure 1. Study overview and research questions. *Left:* zero-shot annotation setup: given an input text and a user-provided task definition, the LLM produces a label prediction. *Right:* we study three facets of the interaction between model-internalized task concepts and user instructions. **RQ1** tests whether task familiarity correlates with performance, contrasting *definition alignment* (Definition-Specific Familiarity; DSF) with *text memorization* metrics (ROUGE-L, BERTScore, embedding similarity; Table 4). **RQ2** measures steerability under an aligned definition by asking whether zero-shot errors can be *rescued* (rescue rate), and how this depends on confidence. **RQ3** probes susceptibility to *misaligned* definitions, testing how performance and confidence change when the provided definition is incorrect. Insets summarize the main empirical findings for each question.

definition.

We investigate this tension through three research questions (Figure 1):

- **RQ1 (Definition Fit and Model Selection):** Does alignment between a model’s internalized concept and a task definition predict which model performs better on that task, beyond text memorization and dataset differences?
- **RQ2 (Decision Stickiness):** To what extent can external knowledge or specialized prompting correct initial zero-shot errors, and how does model confidence relate to correctability?
- **RQ3 (Misalignment Susceptibility):** How do LLMs behave when given misaligned or incorrect task definitions, and can confidence scores detect such misalignment?

We evaluate nine LLMs on five primary toxicity datasets spanning social media, gaming, news, and forums, with three supplementary datasets providing targeted robustness and non-toxicity generalization checks. We make three contributions. First, we introduce *Definition-Specific Familiarity* (DSF), which compares a model’s elicited concept with a dataset’s definition. Across 54 model–dataset pairs, DSF predicts which models perform better within the same task after controlling for dataset identity (partial $r = +0.41$); three text-memorization measures provide no predictive pos-

itive signal. Second, we show that prompting is a limited correction mechanism: only 34.8% of zero-shot errors are rescued by definitions, examples, or automated prompt refinement, and high-confidence errors are especially sticky. Third, misaligned definitions systematically redirect predictions without a corresponding confidence decrease, making confidence unreliable for detecting definition–policy mismatch. Together, these results show that model selection and policy validation cannot be replaced by better prompting or confidence thresholds alone. They extend work on steerability in generation (Chang et al., 2026) by characterizing the limits of prompting as a control channel for discrete annotation decisions. Code and analysis pipelines are available at <https://github.com/etmaca5/1lm-internalized-priors-for-annotation>.

2. Background and Related Work

LLM Steerability and Controllability. Recent work has formalized steerability as the degree to which a model’s behavior can be shifted from its baseline via interventions such as prompting (Miehling et al., 2025). Chang et al. (2026) introduce a multi-dimensional goal-space framework for measuring steerability in text generation, identifying three key failure modes: poor coverage (rare goals under-represented), miscalibration (overshooting requests), and

side effects (unintended changes to unspecified dimensions). Their evaluation reveals that current LLMs struggle with reliable steerability even under various interventions including prompt engineering and reinforcement learning. Our work addresses a complementary question: in *annotation* tasks where the output is a discrete label rather than generated text, what determines whether a model can be steered toward a specific task definition? We find analogous failure modes: limited reachability (stubborn errors), miscalibration (flat confidence under misalignment), and side effects (rescue-corruption tradeoffs).

LLMs as Annotators and Judges. The use of LLMs for evaluating other models or annotating datasets has become a standard paradigm. Gilardi et al. (2023) showed that ChatGPT can outperform crowd-workers on text-annotation tasks, while Zheng et al. (2023) established LLM-as-a-judge evaluation with MT-Bench. Kim et al. (2024) introduced Prometheus, a model specifically fine-tuned for fine-grained rubric-based evaluation, highlighting the importance of specialized alignment for assessment tasks. However, Baumann et al. (2025) replicated 37 annotation tasks from 21 published studies using 18 different language models, finding that configuration choices can lead to incorrect conclusions in 31-50% of cases. Relatedly, Kocielnik et al. (2025a) show that aligning a model’s concept with a curated human definition is a prerequisite for reliable large-scale annotation of player chat, motivating explicit definition-alignment diagnostics. With just a few prompt paraphrases across multiple LLMs, researchers can present most hypotheses as statistically significant - a phenomenon coined “LLM hacking.” This suggests that LLMs should be treated as complex instruments requiring rigorous validation rather than convenient black-box annotators.

Model-Internalized Priors and Annotation Reliability. Modern LLMs develop strong internal priors across the full training pipeline (pre-training, instruction tuning, and RLHF/DPO) that can conflict with user-provided definitions. Carlini et al. (2021) demonstrate that LLMs memorize specific training examples, suggesting models have ingrained conceptions of common concepts like toxicity. Post-training alignment can amplify these effects: Zhang et al. (2026) identify *typicality bias* where annotators systematically favor familiar text during RLHF, causing models to anchor to “typical” examples rather than following nuanced definitions. Min et al. (2022) show that in-context learning is often driven by format and label space rather than input-label mappings, suggesting models may not fully override their priors even with explicit examples. Prior work on benchmark contamination has developed metrics like Min-K% Prob (Shi et al., 2024), BERTScore (Zhang et al., 2020), and continuation-based detection (Golchin & Surdeanu, 2024), but less is known about how internalized *conceptual* priors

(vs. text memorization) constrain steerability.

Confidence and Calibration. Model accuracy alone can mask calibration failures critical for annotation reliability. Surveys of LLM confidence estimation (Geng et al., 2024) highlight challenges with overconfidence and unreliable uncertainty quantification. Black-box confidence elicitation methods reveal that current LLMs lack an internally coherent sense of confidence (Han et al., 2025), often overstating self-reported scores while showing wide variability when prompted to reconsider (Xiong et al., 2024; Pawitan & Holmes, 2025). While verbalized confidence can be better calibrated than token probabilities under RLHF (Tian et al., 2023), prompt design strongly shapes reliability (Yang et al., 2024). We adopt verbalized confidence to enable applicability to closed-source models that do not expose token-level probability scores.

3. Methods

We propose a framework to evaluate the interplay between the priors an LLM has internalized through training and tuning (familiarity) and its steerability via prompting (alignment) in the context of annotation tasks.

3.1. Familiarity Metrics.

We distinguish between two types of familiarity that could influence annotation performance: *text familiarity* (whether the model has memorized specific texts from the dataset) and *definition familiarity* (whether the model’s internal concept aligns with the task definition). These represent different mechanisms: text familiarity suggests performance via generation of memorized content, while definition familiarity suggests performance via conceptual task alignment.

Text Familiarity. Measures memorization by prompting the model to generate a continuation given a prefix comprising 40% of the text instance, then computing **ROUGE-L F1** (based on longest common subsequence) (Lin, 2004) between the generated continuation and the remaining ground truth text. This adapts the continuation-based contamination detection approach of Golchin & Surdeanu (2024) for continuous familiarity measurement: rather than using dataset-specific prompts for binary contamination detection, we use generic continuation prompts (see Appendix B) at temperatures 0.0 and 0.7, taking the maximum score. To ensure findings are robust to the choice of memorization proxy, we additionally compute **BERTScore F1** (Zhang et al., 2020) using DeBERTa-XL contextual token embeddings, and **embedding cosine similarity** using the same sentence encoders as DSF—so any difference between metrics reflects signal type (lexical vs. contextual vs. semantic reproduction) rather than encoder choice. High scores indicate that the model

can reproduce text that it likely encountered during training.

Definition-Specific Familiarity (DSF). Quantifies alignment between a model’s internal understanding of the target phenomenon (e.g., what constitutes toxicity) and the dataset’s operational definition of that phenomenon. The intuition is analogous to human annotators: two annotators given the same definition may interpret it differently based on their prior conceptions and identities (Sap et al., 2022; Plank, 2022). An annotator whose mental model aligns with the definition will perform better than one with a mismatched conception, regardless of whether they have seen the specific texts before.

To compute DSF, we prompt the model to explain its understanding of the target concept (e.g., “In your own words, what makes content toxic?”) and measure semantic similarity between this explanation and the dataset’s full definition using sentence embeddings. To remove sensitivity to any single embedding model, we compute DSF as the unweighted mean (*consensus DSF*) across six diverse sentence encoders: MiniLM, MP-Net, BGE-large, E5-large, Instructor-large, and OpenAI’s `text-embedding-3-small`. Appendix E.1 shows that all six encoders produce positive partial correlations with accuracy and that pairwise DSF values are highly concordant ($r \geq 0.88$). DSF–accuracy correlations are computed across all nine models and six datasets (adding the Jigsaw Unintended Bias dataset (Borkan et al., 2019) to the five primary datasets), yielding $N = 54$ model–dataset pairs. Crucially, DSF can be measured using only the task definition and a handful of prompts, without requiring labeled data or full annotation runs, making it a practical diagnostic for selecting model–task pairings. This quantification captures the entire concept boundary including both positive and negative criteria. We also tested alternative variants incorporating domain-specific prompts and separate positive/negative class definitions, but these showed weaker correlation with model performance on a given task (see Appendix E).

3.2. Steerability Metrics

To quantify an LLM’s ability to correct its errors when provided with better instructions (steerability), we define the **Rescue Rate**:

$$\text{Rescue Rate} = P(\text{Correct} \mid \text{Prompted}, \text{Zero-Shot Wrong}) \quad (1)$$

This metric captures the reachability of correct behaviors via prompting. A low rescue rate indicates that many behaviors are unreachable through prompt-based steering alone. We define **decision stickiness** as the tendency for high-confidence errors to resist correction.

To investigate how LLMs respond to incorrect instructions

```
[Base classification prompt...]
Respond in exactly this format:
PREDICTION: [LABEL/not LABEL]
CONFIDENCE: [0-100]
```

Figure 2. Confidence elicitation prompt suffix appended to all classification prompts. LABEL is replaced with the dataset-specific positive class (e.g., `toxic`, `hateful`, `offensive`, etc.). The base prompt varies by condition (zero-shot, definition, few-shot, misaligned, etc.). This template is identical across all models.

(RQ3), we employ *concept substitution*: providing a definition from a different dataset that operationalizes a related but semantically distinct concept. For example, when annotating a general toxicity dataset, we substitute definitions of hate speech (which requires identity-based targeting) for gaming toxicity (which includes any disruptive behavior). We test 6 misaligned conditions varying in *definition scope*: narrow definitions like hate speech require identity-based targeting, while broad definitions like gaming toxicity capture any disruptive behavior. Full definitions are in Appendix A.

We measure four complementary metrics that decompose how prompting changes behavior relative to zero-shot: **Change Rate** (fraction of flipped labels) captures how responsive a model is to the provided definition. **Rescue Rate** ($P(\text{correct} \mid \text{zero-shot wrong})$) quantifies how often prompting fixes errors. **Corruption Rate** ($P(\text{wrong} \mid \text{zero-shot correct})$) measures how often prompting breaks correct predictions. Models that are easily steered can be steered in both beneficial and harmful directions. Finally, **Prediction Bias** (predicted positive rate - true positive rate) captures whether prompting shifts predictions toward over-calling (more false positives) or under-calling (more false negatives) relative to the dataset’s base rate.

Confidence. We elicit verbalized confidence scores by prompting the model to output a numerical score (0–100) alongside its prediction. The exact prompt suffix is shown in Figure 2. We use the same template across all models; only the label string varies by dataset. All calls use temperature 0.

3.3. Experimental Setup

Models. We evaluate nine instruction-tuned models spanning open-weights and proprietary systems (Table 2), organized in two tiers. Six models form the *core* of our analysis, spanning dense architectures (7B–70B parameters) and mixture-of-experts architectures (Mixtral-8x7B, DeepSeek-V3) across the Llama and Mistral families. All mixed-effects regressions (RQ2, RQ3) are scoped to these six models to keep statistical inference consistent. We additionally evaluate three *extended* models—GPT-4o-mini, Llama-3.3-70B, and Qwen-2.5-72B—under every prompting condition on every primary dataset. These verify that our headline accuracy, rescue-rate, and DSF findings extend to a

Table 1. Five primary toxicity datasets followed by three supplementary datasets ([†]). % Pos. is the proportion of the task-positive label in the original dataset.

Dataset	Domain	Size	% Pos.
Twitter Hate (Davidson et al., 2017)	Social Media	24,783	5.8
OLID (Zampieri et al., 2019)	Social Media	14,100	33.2
GameTox (Naseem et al., 2025)	Gaming	53,000	42.5
Fox News (Gao & Huang, 2017)	News	1,528	28.5
Jigsaw Toxic Comments (cjadams et al., 2018)	Forum	159,571	9.6
Jigsaw Unintended Bias [†] (Borkan et al., 2019)	Forum	97,320	8.0
SemEval-2018 Irony [†] (Van Hee et al., 2018)	Social Media	4,618	48.1
Subjectivity [†] (Pang & Lee, 2004)	Reviews	10,000	50.0

Table 2. Overview of models evaluated. Act. = parameters active per forward pass (MoE models). N/D = not disclosed. *Estimated from unofficial sources; not officially disclosed by OpenAI.

Model	Arch.	Param.	Active
Llama-3.1-8B (Grattafiori et al., 2024)	Dense	8B	8B
Llama-3.1-70B (Grattafiori et al., 2024)	Dense	70B	70B
Llama-3.3-70B (Meta AI, 2024)	Dense	70B	70B
Mistral-7B (Jiang et al., 2023)	Dense	7B	7B
Mistral-Small-24B (Mistral AI, 2025)	Dense	24B	24B
Mixtral-8x7B (Jiang et al., 2024)	MoE	47B	13B
DeepSeek-V3 (DeepSeek-AI, 2024)	MoE	671B	37B
GPT-4o-mini (OpenAI, 2024)	Prop.	8B*	8B*
Qwen-2.5-72B (Qwen et al., 2025)	Dense	72B	72B

broader, more recent slice of proprietary and open-weights systems. For MoE models, we report both total and active parameters per forward pass. Appendix G details which models contribute to each table and figure.

Datasets. We focus on toxicity and hate speech detection across five primary datasets covering a range of online communication environments, from formal news comments to informal gaming chats, each with a different operational definition of toxicity (Table 1). These five form the analytic core and appear in every results table in the main paper. We additionally use three supplementary datasets for targeted robustness and generalization checks: *Jigsaw Unintended Bias*, which extends the DSF panel to $N = 54$ model-dataset pairs and serves as a label-bias robustness check for the rescue-rate analysis; and *SemEval-2018 Irony* together with *Subjectivity*, which provide non-toxicity generalization checks for RQ1 and RQ2 (Appendix F). Appendix G maps each dataset to the specific analyses it contributes to.

Experiments. We evaluate five prompting conditions: (1) Zero-Shot: task name only with no additional context, (2)

Aligned Definition: task name plus the dataset’s official definition of the target concept, (3) Few-Shot: task name plus 4 random balanced examples, (4) FS+Def: the aligned definition combined with 4 few-shot examples, and (5) Misaligned: definitions from other datasets applied to the target dataset (6 variants, detailed in Appendix A). All prompts follow a consistent template where the classification question is followed by the message and an “Answer:” suffix. We additionally run two DSPy (Khatab et al., 2023) automated optimization conditions on all nine models: DSPy Optimized (automatically selected few-shot examples) and DSPy Aligned (selected examples plus aligned definition). For each dataset-model pair, we randomly sampled 1,000 instances. All inference was conducted via API at temperature 0 for reproducibility.

Statistical Analysis For RQ1, we examined associations between familiarity metrics and annotation accuracy using regression with cluster-robust standard errors (to account for dependence within models) and partial correlations controlling for dataset. For RQ2 and RQ3, we used mixed effects logistic regression on the six open-weights models to account for repeated observations of the same texts across conditions. This handles the binary outcome (correct vs. incorrect) while modeling the non-independence of the same text being evaluated under multiple prompting conditions. We include random intercepts for text and fixed effects for model, prompting condition, domain, and their interactions.

4. Results

Our core analysis covers predictions across 9 models, 5 toxicity datasets, and 10 prompting conditions, plus two DSPy automated optimization conditions. Table 3 summarizes model performance and steerability, while Table 5 presents key findings from our mixed effects regression models.

4.1. Answering RQ1: Impact of Task Familiarity

We analyzed whether annotation performance correlates more strongly with text memorization or semantic task-definition alignment (Definition-Specific Familiarity, DSF). The unit of analysis is a model-dataset cell: text-reproduction scores are first averaged over examples within each cell, just as accuracy is, while DSF is measured for that model and definition. We then partial out dataset identity. The resulting coefficient therefore answers a within-task question: after removing average differences among datasets, do models whose elicited concept is closer to a dataset’s definition perform better on that same task?

In zero-shot evaluation, consensus DSF is positively associated with accuracy (partial $r = +0.41$, 95% CI $[+0.15, +0.62]$, $p = 0.003$, $N = 54$ model-dataset cells). This is not an artifact of one sentence encoder: the six encoder-specific estimates are MiniLM +.36, MPNet +.33,

Table 3. Model performance (Accuracy % / AUC) across prompting conditions. Misaligned averages 6 definition-swap conditions (full breakdown in Appendix D).

Model	Zero-Shot	Aligned Def	Few-Shot	FS + Def	DSPy Opt.	DSPy Aligned	Misaligned	Overall
Llama-3.1-8B	76.7/.832	80.1/.843	75.3/.822	75.4/.824	72.3 [†] /.809	73.5/.821	77.0/.822	76.8/.818
Llama-3.1-70B	79.8/.881	82.1/.888	81.2/.892	82.1/.898	79.9 / .889	82.6/.899	78.1/.859	79.7/.869
Llama-3.3-70B	80.5/ .885	82.3/ .892	82.0/ .896	83.0/ .904	77.5 [†] /.884	81.3/ .904	79.0/.863	80.1/ .874
Mistral-7B	80.5/.858	81.3/.787	81.5/.852	80.8/.838	82.0 /.827	81.7/.825	81.4 /.823	81.2/.819
Mistral-Small-24B	78.0/.868	81.0/.863	80.8/.870	82.3/.879	79.3 /.867	81.2/.876	80.7/.850	80.5/.852
Mixtral-8x7B	81.0/.868	82.7/.868	82.4/.868	80.9/.859	82.8 /.860	82.7/.861	81.2/.843	81.6/.850
DeepSeek-V3	81.3/.874	83.0/.881	82.6/.890	83.8 /.893	80.9 /.877	83.3/.890	80.7/.856	81.6/.865
GPT-4o-mini	81.6/.871	83.3 /.875	84.1 /.880	83.3/.880	84.4 /.883	84.3 /.883	81.1/.857	82.3 /.862
Qwen-2.5-72B	83.3 /.876	82.2/.879	83.8/.894	83.2/.892	82.5 /.887	83.2/.891	81.2/ .868	82.2/.872
Cond. Avg	80.3/.868	82.0/.864	81.5/.874	81.6/.874	80.2 /.865	81.5/.872	80.0/.849	80.8/.853

[†]DSPy Opt. degrades Llama-3.1-8B (76.7→72.3) and Llama-3.3-70B (80.5→77.5), consistent with Tan et al. (2025).

BGE-large +.49, E5-large +.43, Instructor-large +.30, and OpenAI +.43 (Appendix E.1). Their DSF score vectors are also highly concordant (pairwise $r = .88-.99$), showing that the encoders largely agree about which model-definition pairs are aligned.

Figure 3 extends this comparison across six prompting conditions. DSF is strongest with few-shot examples (partial $r = +0.62$, 95% CI [+0.41, +0.77]) and few-shot examples plus the definition ($r = +0.53$, [+0.30, +0.71]). It remains positive in the other conditions: DSPy prompts ($r = +0.24$ without and +0.37 with the definition), zero-shot, and definition-only prompting. The three per-text memorization measures show a different pattern: after controlling for dataset, their correlations with whether an individual text is classified correctly are small and generally non-positive across conditions. The consistently positive DSF correlations show that DSF captures meaningful within-task differences between models, identifying which models are better suited to a given task rather than merely reflecting dataset difficulty irrespective of prompting technique used.

Table 4 contrasts this conceptual-alignment signal with three text-reproduction measures in the zero-shot condition. Their strong raw correlations largely reflect between-dataset differences: Jigsaw, for example, is both relatively high-memorization and difficult. Once dataset identity is controlled for, ROUGE-L, BERTScore, and embedding similarity show no positive association with performance (partial $r = -.19, -.15, -.16$, respectively), whereas consensus DSF remains positive. *Models whose elicited concepts more closely match a dataset’s definition tend to perform better on that task, while text memorization does not provide the same signal.*

The same positive association appears in supplementary experiments on irony and subjectivity detection (partial $r = +0.34$, $N = 18$ model-dataset pairs), providing directional evidence that the DSF result is not confined to toxicity; see Appendix F.

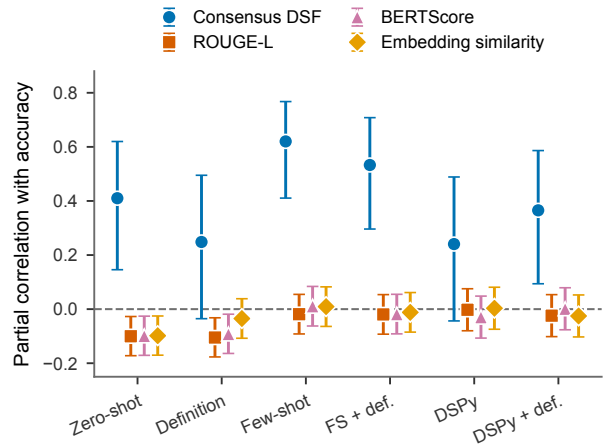


Figure 3. Partial correlations controlling for dataset. DSF uses cell-level accuracy ($N = 54$ per condition); memorization metrics use per-text correctness ($N = 722$; DSPy: $N = 641$). Bars show 95% CIs.

Table 4. Correlations between familiarity metrics and zero-shot annotation accuracy on the extended $N = 54$ panel. Raw correlations are misleading due to dataset confounds; partial correlations (controlling for dataset) reveal that definition alignment (consensus DSF) is positively associated with performance while three distinct memorization metrics are not.

Metric	Raw r	Partial r
Text Memorization (ROUGE-L)	-0.80	-0.19
Memorization (BERTScore)	-0.76	-0.15
Memorization (Embedding Similarity)	-0.71	-0.16
Definition Familiarity (consensus DSF)	+0.74	+0.41

4.2. Answering RQ2: Impact of Knowledge Injection

Table 5 presents the factors associated with both annotation correctness and error correction, evaluated via mixed-effects logistic regression. Domain is most strongly associated with correctness: compared with Forum, the odds are approximately 14× higher for Gaming ($OR = 13.85$) and 5× higher for Social Media ($OR = 5.45$). Aligned definitions improve mean accuracy by only 2.2 points over zero-shot,

and DSPy changes the corresponding averages by at most 0.5 points (Table 3). Moreover, DSPy degrades both Llama models, consistent with prior findings that prompt optimizers can underperform unoptimized baselines (Tan et al., 2025). These modest, heterogeneous gains suggest that suboptimal prompt design alone does not explain the performance ceiling. Conditional on other covariates, a correct zero-shot response predicts $6.4\times$ higher odds of correctness with definitions or examples ($OR = 6.43$), suggesting that prompting consolidates correct answers more readily than it rescues errors.

Table 5. Key findings from mixed effects logistic regression. $OR > 1$ indicates higher odds of a correct annotation. Reference categories: Forum (domain), Llama-8B (model). Full model specifications in Appendix C. All $p < 0.001$.

Factor	OR [95% CI]
<i>Factors Associated with Correctness:</i>	
Gaming (vs. Forum)	13.85 [13.31, 14.41]
Social Media (vs. Forum)	5.45 [5.27, 5.63]
Aligned Definition	1.08 [1.04, 1.11]
Aligned $\times$ News	1.91 [1.80, 2.02]
Zero-shot Correct	6.43 [6.28, 6.58]
<i>Factors Associated with Error Correction:</i>	
ZS Confidence (per SD)	0.84 [0.82, 0.87]

Table 6. Rescue rate by model, aggregated across prompted conditions. Rescue rate is $P(\text{correct} \mid \text{prompted, zero-shot wrong})$.

Model	Rescue Rate	95% CI	N (ZS err.)
Mistral-small-24B	44.2%	[43.1, 45.2]	8,536
DeepSeek-V3	38.1%	[36.9, 39.2]	7,264
Llama-3.1-8B	37.5%	[36.5, 38.5]	9,040
GPT-4o-mini	35.7%	[34.6, 36.8]	7,152
Mixtral-8x7B	34.6%	[33.5, 35.7]	7,416
Mistral-7B	31.6%	[30.5, 32.6]	7,568
Llama-3.1-70B	31.3%	[30.3, 32.4]	7,848
Llama-3.3-70B	30.4%	[29.4, 31.4]	7,592
Qwen-2.5-72b	27.8%	[26.8, 28.9]	6,504

Rescue Rate is the probability that prompting corrects a zero-shot error. The overall rescue rate is only 34.8%, meaning nearly two-thirds of errors resist correction. The same qualitative pattern appears on irony and subjectivity: few-shot prompting rescues 45.0% of 3,052 zero-shot errors, leaving most errors uncorrected (Appendix F). Figure 4 shows an inverted-U: rescue probability peaks at 51.8% for moderate confidence (0.6–0.7) and falls sharply to 20.8% for high-confidence errors (> 0.9), the most statistically reliable region of the curve. Model 4 corroborates this trend: each standard-deviation increase in zero-shot confidence reduces rescue odds by 16% ($OR = 0.84$, Table 5). *When models are confidently wrong, prompting rarely helps.*

The drop at the extreme low-confidence end ($n < 1,000$) is a distinct failure mode rather than a counterexample to decision stickiness. These instances are out-of-distribution and context-deficient: their ROUGE-L is far below the rest of

the data (0.025 vs. 0.061) and they are dramatically shorter (median 6 vs. 22 words), consisting largely of single-word gaming fragments (e.g., “noob”, “wtf”, “bot”) whose toxicity depends on context that is unavailable to the model.

Importantly, steerability is not a monotonic function of model size: for example, Mistral-Small-24B rescues more zero-shot errors than Llama-3.1-70B in our experiments, despite being substantially smaller (Table 6). To confirm this finding is not an artifact of label-bias issues known to affect the standard Jigsaw dataset (e.g., systematic flagging of African American Vernacular English as toxic), we replicated the rescue rate analysis on the Jigsaw Unintended Bias dataset (Borkan et al., 2019), which was explicitly designed to reduce identity-group annotation bias. Rescue rates remain uniformly below 50% across all nine models (Table 18), confirming that decision stickiness is not specific to Jigsaw’s label properties.

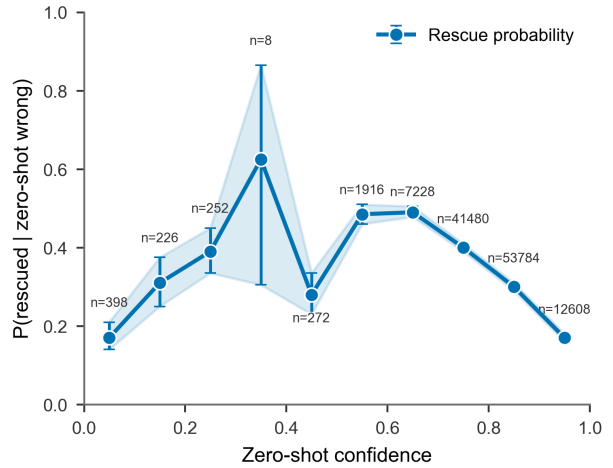


Figure 4. Rescue probability vs. zero-shot confidence for zero-shot errors. The inverted-U shape exhibits two distinct failure modes (analyzed in *Confidence and Decision Stickiness*): *decision stickiness* at high confidence (right tail) and an out-of-distribution effect at very low confidence (left tail).

4.3. Answering RQ3: Misalignment Impact

We analyze 6 misaligned definition conditions to investigate how LLMs respond to incorrect instructions (Table 7). The results reveal several surprising patterns.

We find that LLMs are responsive to definition scope. Narrow definitions (requiring targeting based on race, religion, gender, etc.) produce prediction bias of -7% to -12% (under-prediction), while broad definitions produce bias of $+9\%$ to $+13\%$ (over-prediction). This shift in prediction rates demonstrates that LLMs do not simply ignore misaligned instructions; they adjust decision threshold to match the provided criteria. This responsiveness is a double-edged sword: while it enables definition-based steering, it also means models will confidently apply incorrect definitions.

The worst misaligned condition (gaming toxicity: 76.4%) is 5.6% below aligned, while the best (Fox News hate speech: 82.6%) actually *outperforms* the aligned definition by 0.6%. Knowledge built up during training and tuning constrains how much definitions can shift behavior: even with “wrong” definitions, models remain relatively stable, indicating persistent influence from these internalized task concepts.

Surprisingly, some misaligned definitions provide net benefit over zero-shot (rescuing more errors than they corrupt). For instance, using a “hate speech” definition can improve performance on general toxicity tasks by focusing the model on clearer criteria, even if technically misaligned. This finding challenges the assumption that the “correct” definition is always optimal - empirically, some misaligned definitions better match the model’s internal concept.

Table 7. Misalignment analysis: LLMs adjust prediction thresholds based on definition, yet maintain high confidence. Rows are ordered by definition breadth (Narrow → Broad). Confidence remains 84–91% regardless of alignment. All values are percentages.

Condition	Acc.	Change	Bias	Conf.
Aligned definition	82.0	11.3	ref	88.1
Fox News hate speech	82.6	17.5	−7.2	90.6
Twitter hate speech	78.5	23.1	−12.1	88.1
GameTox toxicity	79.6	15.4	+0.5	86.5
Olid offensive	81.9	10.9	+5.1	89.1
General toxicity	81.1	9.2	+8.8	85.0
Gaming toxicity	76.4	13.3	+13.1	88.5

Critical Calibration Failure. Models *remain confident even when applying misaligned definitions* (Table 7): mean confidence changes little from zero-shot (87.0%) across aligned and misaligned definitions (85.0–90.6%). The narrowest definition (Fox News hate speech) produces the *highest* confidence, not the lowest. This represents a calibration failure: models should exhibit increased uncertainty when instructions contradict their training distribution, but they do not. As shown in Figure 5, all conditions (zero-shot, aligned, and misaligned) exhibit similar overconfidence patterns, with calibration curves falling below the diagonal. Critically, there is no meaningful separation between aligned and misaligned definitions, meaning *practitioners cannot rely on confidence scores to detect definition errors*.

Across all models and datasets, definition choice produces 17% accuracy variation (e.g., for Fox News, accuracy ranges from 61.6% with the “Gaming Toxicity” definition to 78.4% with the “Twitter Hate Speech” definition), while model choice produces only 5% variation on average. This implies that for annotation tasks, *careful definition design is more impactful than selecting larger or more capable models*.

Models showing high rescue rates under misalignment also show high corruption rates (Table 6 and Table 17). Mistral-Small-24B achieves the highest rescue rate (73.7% with

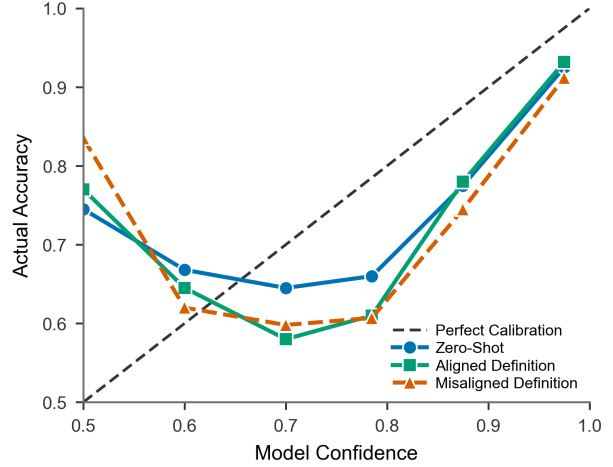


Figure 5. Calibration curves showing confidence vs. actual accuracy across conditions. All conditions exhibit overconfidence (below the diagonal), with no meaningful separation between aligned and misaligned definitions: models cannot distinguish when they are applying incorrect instructions.

Twitter hate speech) but also the highest corruption rate (21.0%). This creates a fundamental trade-off: *models that are easily steered by definitions can be steered in both beneficial and harmful directions*.

5. Discussion

DSF predicts accuracy after accounting for dataset differences, whereas three text-memorization metrics do not. This supports a concept-level account of annotation performance and shifts validation from auditing text overlap toward assessing whether a model’s concept boundary matches the intended definition.

Prompting has limited ability to overturn wrong decisions. We find substantial *decision stickiness*: most zero-shot errors persist under aligned definitions and few-shot examples, and high-confidence errors are especially resistant to correction (Figure 4, Table 5). This indicates that prompting primarily stabilizes or modestly improves predictions the model already gets right, rather than serving as a reliable mechanism for error recovery. We further test whether iterative, history-aware prompting can break this *stickiness*: a three-turn rescue sequence that progressively adds few-shot examples, the aligned definition, and an explicit reconsideration prompt raises rescue from 7.5% at Turn 1 to only 18.7% at Turn 3, and lifts high-confidence rescue to just 8.5% (Appendix D.1). Decision stickiness therefore persists under iterative correction, not only one-shot prompting. Notably, steerability is not monotonic in model size (Table 6), implying that larger models are not necessarily easier to steer for annotation workloads.

Misaligned definitions shift behavior without reliably signaling risk. When given semantically related but mis-

matched definitions, models change their prediction rates in systematic ways (Table 7, see also Table 15 in Appendix), consistent with literal application of the provided criteria. However, accuracy often degrades only modestly and sometimes even improves relative to the aligned condition. One interpretation is that the model’s training supplies a strong prior that anchors behavior, providing a performance floor even under imperfect instructions. Another possibility is that dataset label boundaries may be closer to some alternative formulations than the dataset’s stated definition, especially for inherently subjective constructs. Regardless, the key operational point is that *definition wording is a first-order experimental factor*: in our study, definition choice can induce substantial performance variation, so treating the definition as fixed can meaningfully understate uncertainty in downstream analyses.

Confidence is conditional, not diagnostic of definition validity. A central deployment risk is that models remain highly confident under misalignment, and calibration curves show little separation between aligned and misaligned conditions (Figure 5). Importantly, this does *not* imply that the model “should know” the definition is wrong; rather, it suggests that the reported confidence reflects certainty *given the provided instructions*, not a meta-uncertainty about whether those instructions match the intended labeling policy. As a result, confidence thresholds are poorly suited for detecting definition errors or definition-dataset mismatch.

Practical implications for LLM annotation pipelines. These findings suggest three lightweight safeguards. First, measure *definition alignment* before large-scale annotation (e.g., DSF-style checks) to anticipate which model–definition pairs are likely to work well. Second, *stress-test definitions* by evaluating multiple plausible formulations and reporting sensitivity (e.g., changes in bias, rescue, and corruption), rather than assuming a single “correct” wording is robust. Third, avoid using confidence as a proxy for definition appropriateness.

Limitations and future work. Our study focuses primarily on binary toxicity classification with concept-substitution misalignment. Supplementary irony and subjectivity analyses extend beyond toxicity, but multi-class, span-labeling, and open-ended tasks may have different failure modes. Consensus DSF across six encoders reduces embedding-choice sensitivity (Appendix E.1), but alternative similarity measures and elicitation prompts need further exploration. Because all models are instruction-tuned, we cannot separate capability limits from intentional safety constraints or identify which training stage produces concept anchoring. Addressing this requires matched base and post-trained models. Results measure fidelity to dataset-specific annotation policies rather than normative correctness, although replication on Jigsaw Unintended Bias and stability across datasets

reduce dependence on any single policy’s ambiguity or bias. Finally, our evidence is correlational: causal claims would require interventions that manipulate DSF while holding other factors fixed. Future work should test stronger forms of misalignment, alternative multi-turn correction strategies such as self-consistency or critique-and-revise, and scalable procedures for selecting model–definition pairs under explicit constraints, all of which present significant trade-offs in deployment cost and complexity even if proven effective.

6. Conclusion

We study how model-internalized priors interact with prompt-based steering in LLM annotation. Across 9 models and 5 toxicity datasets, we find that performance is best explained by *definition-specific familiarity* (DSF): controlling for dataset difficulty, DSF is positively associated with accuracy (partial $r = +0.41$, $N = 54$), while three distinct memorization metrics (ROUGE-L, BERTScore, embedding similarity) are not. Prompting often fails to correct mistakes, as most zero-shot errors persist even under aligned definitions and few-shot examples. Models can confidently follow *misaligned* definitions, making confidence an unreliable indicator of definition correctness. Finally, definition wording induces larger performance swings than model choice, highlighting definition design as crucial. Practically, LLM annotation pipelines should prioritize measuring definition alignment as an early indicator of performance rather than relying on larger models or confidence-based strategies.

Acknowledgment

This work is supported by the Caltech Linde Center for Science, Society, and Public Policy (LCSSP). R. Michael Alvarez is Flintridge Foundation Professor of Political and Computational Social Science at Caltech. We thank Andrea Boonyarungsrit, Grant Cahill, Jonathan Lane, Min Kim, and Fereshteh Soltani for prior collaboration that helped inform some of the motivating challenges behind this work. The views and opinions expressed in this paper are solely those of the authors and do not necessarily reflect those of the individuals acknowledged or their affiliated organizations.

Impact Statement

This work examines how model-internalized priors interact with prompt-based steering in LLM annotation tasks. Our findings have implications for practitioners deploying LLMs as annotators or judges. Positively, identifying failure modes such as decision stickiness and miscalibration under definition misalignment can help practitioners develop better validation procedures and avoid over-reliance on confidence scores. However, our results also reveal that models will confidently apply incorrect definitions when provided, which could lead to systematic errors if task specifications are poorly designed.

References

- Baumann, J., Röttger, P., Urman, A., Wendsjö, A., Plaza-del Arco, F. M., Gruber, J. B., and Hovy, D. Large language model hacking: Quantifying the hidden risks of using LLMs for text annotation. *arXiv preprint arXiv:2509.08825*, 2025. doi: 10.48550/arXiv.2509.08825. URL <https://arxiv.org/abs/2509.08825>.
- Borkan, D., Dixon, L., Sorensen, J., Thain, N., and Vasserman, L. Nuanced metrics for measuring unintended bias with real data for text classification. In *Companion Proceedings of The 2019 World Wide Web Conference*, WWW '19, pp. 491–500, New York, NY, USA, 2019. Association for Computing Machinery. ISBN 9781450366755. doi: 10.1145/3308560.3317593. URL <https://doi.org/10.1145/3308560.3317593>.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., and Amodei, D. Language models are few-shot learners. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H. (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 1877–1901. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcbb4967418bfb8ac142f64a-Paper.pdf.
- Carlini, N., Tramèr, F., Wallace, E., Jagielski, M., Herbert-Voss, A., Lee, K., Roberts, A., Brown, T., Song, D., Erlingsson, Ú., Oprea, A., and Raffel, C. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pp. 2633–2650. USENIX Association, August 2021. ISBN 978-1-939133-24-3. URL <https://www.usenix.org/conference/usenixsecurity21/presentation/carlini-extracting>.
- Chang, T., Schnabel, T., Swaminathan, A., and Wiens, J. A course correction in steerability evaluation: Revealing miscalibration and side effects in LLMs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. URL <https://arxiv.org/abs/2505.23816>.
- cjadams, Sorensen, J., Elliott, J., Dixon, L., McDonald, M., nithum, and Cukierski, W. Toxic comment classification challenge. Kaggle Competition, 2018. URL <https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge>.
- Davidson, T., Warmley, D., Macy, M., and Weber, I. Automated hate speech detection and the problem of offensive language. *Proceedings of the International AAAI Conference on Web and Social Media*, 11(1):512–515, 2017. doi: 10.1609/icwsm.v11i1.14955.
- DeepSeek-AI. Deepseek-v3 technical report, 2024. URL <https://arxiv.org/abs/2412.19437>.
- ElSherief, M., Kulkarni, V., Nguyen, D., Wang, W. Y., and Belding, E. Hate lingo: A target-based linguistic analysis of hate speech in social media. *Proceedings of the International AAAI Conference on Web and Social Media*, 12(1), June 2018. doi: 10.1609/icwsm.v12i1.15041. URL <https://ojs.aaai.org/index.php/ICWSM/article/view/15041>.
- Gao, L. and Huang, R. Detecting online hate speech using context aware models. In Mitkov, R. and Angelova, G. (eds.), *Proceedings of the International Conference Recent Advances in Natural Language Processing, RANLP 2017*, pp. 260–266, Varna, Bulgaria, September 2017. INCOMA Ltd. doi: 10.26615/978-954-452-049-6.036. URL <https://aclanthology.org/R17-1036/>.
- Geng, J., Cai, F., Wang, Y., Koeppl, H., Nakov, P., and Gurevych, I. A survey of confidence estimation and calibration in large language models. In Duh, K., Gomez, H., and Bethard, S. (eds.), *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 6577–6595, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.366. URL <https://aclanthology.org/2024.naacl-long.366/>.
- Gilardi, F., Alizadeh, M., and Kubli, M. Chatgpt outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30): e2305016120, 2023. doi: 10.1073/pnas.2305016120. URL <https://www.pnas.org/doi/abs/10.1073/pnas.2305016120>.
- Golchin, S. and Surdeanu, M. Time travel in LLMs: Tracing data contamination in large language models. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=2Rwq6c3tvr>.
- Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.

- Han, P., Kocielnik, R., Song, P., Debnath, R., Mobbs, D., Anandkumar, A., and Alvarez, R. M. The personality illusion: Revealing dissociation between self-reports & behavior in LLMs. *arXiv preprint arXiv:2509.03730*, 2025. URL <https://arxiv.org/abs/2509.03730>.
- Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- Jiang, A. Q., Sablayrolles, A., Roux, A., Mensch, A., Savary, B., Bamford, C., Chaplot, D. S., de las Casas, D., Bou Hanna, E., Bressand, F., Lengyel, G., Bour, G., Lample, G., Lavaud, L. R., Saulnier, L., Lachaux, M.-A., Stock, P., Subramanian, S., Yang, S., Antoniak, S., Le Scao, T., Gervet, T., Lavril, T., Wang, T., Lacroix, T., and El Sayed, W. Mixtral of experts. *arXiv preprint arXiv:2401.04088*, 2024. URL <https://arxiv.org/abs/2401.04088>.
- Khattab, O., Singhvi, A., Maheshwari, P., Zhang, Z., Santhanam, K., Vardhamanan, S., Haq, S., Sharma, A., Joshi, T. T., Moazam, H., Miller, H., Zaharia, M., and Potts, C. Dspy: Compiling declarative language model calls into self-improving pipelines. *ArXiv*, abs/2310.03714, 2023. URL <https://api.semanticscholar.org/CorpusID:263671701>.
- Kim, S., Shin, J., Cho, Y., Jang, J., Longpre, S., Lee, H., Yun, S., Shin, S., Kim, S., Thorne, J., and Seo, M. Prometheus: Inducing fine-grained evaluation capability in language models. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=8euJaTveKw>.
- Kocielnik, R., Li, Z., Kann, C., Sambrano, D., Morrier, J., Linegar, M., Taylor, C., Kim, M., Naqvie, N., Soltani, F., Dehpanah, A., Cahill, G., Anandkumar, A., and Alvarez, R. M. Challenges in moderating disruptive player behavior in online competitive action games. *Frontiers in Computer Science*, 6:1283735, 2024. doi: 10.3389/fcomp.2024.1283735. URL <https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2024.1283735/full>.
- Kocielnik, R., Kim, M., Boonyarungsrit, P. A., Soltani, F., Sambrano, D., Anandkumar, A., and Alvarez, R. M. Prosocial behavior detection in player game chat: From aligning human-AI definitions to efficient annotation at scale. *arXiv preprint arXiv:2508.05938*, 2025a. URL <https://arxiv.org/abs/2508.05938>.
- Kocielnik, R., Li, Z., Linegar, M., Sambrano, D., Soltani, F., Kim, M., Naqvie, N., Cahill, G., Anandkumar, A., and Alvarez, R. M. Online moderation in competitive action games: How intervention affects player behaviors. *Proc. ACM Hum.-Comput. Interact.*, 9(6), October 2025b. doi: 10.1145/3748599. URL <https://doi.org/10.1145/3748599>.
- Lin, C.-Y. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pp. 74–81, Barcelona, Spain, July 2004. Association for Computational Linguistics. URL <https://aclanthology.org/W04-1013/>.
- Liu, Y., Iter, D., Xu, Y., Wang, S., Xu, R., and Zhu, C. G-eval: NLG evaluation using gpt-4 with better human alignment. In Bouamor, H., Pino, J., and Bali, K. (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 2511–2522, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.153. URL <https://aclanthology.org/2023.emnlp-main.153/>.
- Meta AI. Llama 3.3 model card. <https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct>, December 2024.
- Miehling, E., Desmond, M., Natesan Ramamurthy, K., Daly, E. M., Varshney, K. R., Farchi, E., Dognin, P., Rios, J., Bouneffouf, D., Liu, M., and Sattigeri, P. Evaluating the prompt steerability of large language models. In Chiruzzo, L., Ritter, A., and Wang, L. (eds.), *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 7874–7900, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.400. URL <https://aclanthology.org/2025.naacl-long.400/>.
- Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In Goldberg, Y., Kozareva, Z., and Zhang, Y. (eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 11048–11064, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.759. URL <https://aclanthology.org/2022.emnlp-main.759/>.
- Mistral AI. Mistral-small-24b-instruct-2501. <https://huggingface.co/mistralai/Mistral-Small-24B-Instruct-2501>, 2025.
- Mohammad, S., Kiritchenko, S., Sobhani, P., Zhu, X., and Cherry, C. SemEval-2016 task 6: Detecting stance

- in tweets. In Bethard, S., Carpuat, M., Cer, D., Jurgens, D., Nakov, P., and Zesch, T. (eds.), *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)*, pp. 31–41, San Diego, California, June 2016. Association for Computational Linguistics. doi: 10.18653/v1/S16-1003. URL <https://aclanthology.org/S16-1003/>.
- Nadeem, M., Bethke, A., and Reddy, S. StereoSet: Measuring stereotypical bias in pretrained language models. In Zong, C., Xia, F., Li, W., and Navigli, R. (eds.), *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 5356–5371, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.416. URL <https://aclanthology.org/2021.acl-long.416/>.
- Nangia, N., Vania, C., Bhalerao, R., and Bowman, S. R. CrowS-pairs: A challenge dataset for measuring social biases in masked language models. In Webber, B., Cohn, T., He, Y., and Liu, Y. (eds.), *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1953–1967, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.154. URL <https://aclanthology.org/2020.emnlp-main.154/>.
- Naseem, U., Shiwakoti, S., Shah, S. B., Thapa, S., and Zhang, Q. GameTox: A comprehensive dataset and analysis for enhanced toxicity detection in online gaming communities. In Chiruzzo, L., Ritter, A., and Wang, L. (eds.), *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, pp. 440–447, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-190-2. doi: 10.18653/v1/2025.naacl-short.37. URL <https://aclanthology.org/2025.naacl-short.37/>.
- OpenAI. GPT-4o mini: Advancing cost-efficient intelligence. OpenAI Blog, 2024. URL <https://openai.com/blog/gpt-4o-mini-advancing-cost-efficient-intelligence>.
- Pang, B. and Lee, L. A sentimental education: Sentiment analysis using subjectivity summarization based on minimum cuts. In *Proceedings of the 42nd Annual Meeting of the Association for Computational Linguistics (ACL-04)*, pp. 271–278, Barcelona, Spain, July 2004. doi: 10.3115/1218955.1218990. URL <https://aclanthology.org/P04-1035/>.
- Parrish, A., Chen, A., Nangia, N., Padmakumar, V., Phang, J., Thompson, J., Htut, P. M., and Bowman, S. R. BBQ: A hand-built bias benchmark for question answering. In Muresan, S., Nakov, P., and Villavicencio, A. (eds.), *Findings of the Association for Computational Linguistics: ACL 2022*, pp. 2086–2105, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.findings-acl.165. URL <https://aclanthology.org/2022.findings-acl.165/>.
- Pawitan, Y. and Holmes, C. Confidence in the Reasoning of Large Language Models. *Harvard Data Science Review*, 7(1), January 2025. <https://hdsr.mitpress.mit.edu/pub/daq0vvpb>.
- Plank, B. The “problem” of human label variation: On ground truth in data, modeling and evaluation. In Goldberg, Y., Kozareva, Z., and Zhang, Y. (eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 10671–10682, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.731. URL <https://aclanthology.org/2022.emnlp-main.731/>.
- Qwen, Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Wei, H., Lin, H., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Lin, J., Dang, K., Lu, K., Bao, K., Yang, K., Yu, L., Li, M., Xue, M., Zhang, P., Zhu, Q., Men, R., Lin, R., Li, T., Tang, T., Xia, T., Ren, X., Ren, X., Fan, Y., Su, Y., Zhang, Y., Wan, Y., Liu, Y., Cui, Z., Zhang, Z., and Qiu, Z. Qwen2.5 Technical Report, January 2025. URL <http://arxiv.org/abs/2412.15115>. arXiv:2412.15115 [cs].
- Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., and Hashimoto, T. Whose opinions do language models reflect? In Krause, A., Brunskill, E., Cho, K., Engelhardt, B., Sabato, S., and Scarlett, J. (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 29971–30004. PMLR, 23–29 Jul 2023. URL <https://proceedings.mlr.press/v202/santurkar23a.html>.
- Sap, M., Swayamdipta, S., Vianna, L., Zhou, X., Choi, Y., and Smith, N. A. Annotators with attitudes: How annotator beliefs and identities bias toxic language detection. In Carpuat, M., de Marneffe, M.-C., and Meza Ruiz, I. V. (eds.), *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 5884–5906, Seattle, United States, July 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.naacl-main.431. URL <https://aclanthology.org/2022.naacl-main.431/>.

- Shaikh, O., Zhang, H., Held, W., Bernstein, M., and Yang, D. On second thought, let’s not think step by step! bias and toxicity in zero-shot reasoning. In Rogers, A., Boyd-Graber, J., and Okazaki, N. (eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 4454–4470, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.244. URL <https://aclanthology.org/2023.acl-long.244/>.
- Shi, W., Ajith, A., Xia, M., Huang, Y., Liu, D., Blevins, T., Chen, D., and Zettlemoyer, L. Detecting pretraining data from large language models. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=zWqr3MQuNs>.
- Tan, S., Agrawal, L. A., Singhvi, A., Lai, L., Ryan, M. J., Klein, D., Khattab, O., Sen, K., and Zaharia, M. LangProBe: a language program benchmark. In Christodoulopoulos, C., Chakraborty, T., Rose, C., and Peng, V. (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 21489–21509, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-335-7. doi: 10.18653/v1/2025.findings-emnlp.1172. URL <https://aclanthology.org/2025.findings-emnlp.1172/>.
- Tian, K., Mitchell, E., Zhou, A., Sharma, A., Rafailov, R., Yao, H., Finn, C., and Manning, C. D. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. pp. 5433–5442, December 2023. doi: 10.18653/v1/2023.emnlp-main.330. URL <https://aclanthology.org/2023.emnlp-main.330/>.
- Van Hee, C., Lefever, E., and Hoste, V. SemEval-2018 task 3: Irony detection in English tweets. In Apidianaki, M., Mohammad, S. M., May, J., Shutova, E., Bethard, S., and Carpuat, M. (eds.), *Proceedings of the 12th International Workshop on Semantic Evaluation*, pp. 39–50, New Orleans, Louisiana, June 2018. Association for Computational Linguistics. doi: 10.18653/v1/S18-1005. URL <https://aclanthology.org/S18-1005/>.
- Weld, H., Huang, G., Lee, J., Zhang, T., Wang, K., Guo, X., Long, S., Poon, J., and Han, C. CONDA: a CONtextual dual-annotated dataset for in-game toxicity understanding and detection. In Zong, C., Xia, F., Li, W., and Navigli, R. (eds.), *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pp. 2406–2416, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.findings-acl.213. URL <https://aclanthology.org/2021.findings-acl.213/>.
- Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., and Hooi, B. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=gjeQKFxFpZ>.
- Yang, D., Tsai, Y.-H. H., and Yamada, M. On verbalized confidence scores for LLMs. *arXiv preprint arXiv:2412.14737*, 2024. URL <https://arxiv.org/abs/2412.14737>.
- Zampieri, M., Malmasi, S., Nakov, P., Rosenthal, S., Farra, N., and Kumar, R. SemEval-2019 task 6: Identifying and categorizing offensive language in social media (OffensEval). In May, J., Shutova, E., Herbelot, A., Zhu, X., Apidianaki, M., and Mohammad, S. M. (eds.), *Proceedings of the 13th International Workshop on Semantic Evaluation*, pp. 75–86, Minneapolis, Minnesota, USA, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/S19-2010. URL <https://aclanthology.org/S19-2010/>.
- Zhang, J., Yu, S., Chong, D., Sicilia, A., Tomz, M., Manning, C. D., and Shi, W. Verbalized sampling: How to mitigate mode collapse and unlock LLM diversity. In *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=9jQkmGunGo>.
- Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., and Artzi, Y. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations (ICLR)*, 2020. URL <https://openreview.net/forum?id=SkeHuCVFDr>.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., Zhang, H., Gonzalez, J. E., and Stoica, I. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023. URL <https://openreview.net/forum?id=uccHPGDlao>.

A. Definition Texts Used

This section provides the exact definition texts given to models, organized by scope (narrow to broad). Understanding definition scope is critical for interpreting misalignment results.

A.1. Narrow Scope Definitions

These definitions require identity-based targeting; general insults without identity targeting are explicitly excluded.

Twitter Hate Speech (Davidson et al., 2017).

Hate Speech: Language that expresses hatred towards a targeted group or is intended to be derogatory, to humiliate, or to insult the members of the group based on attributes such as race, religion, ethnic origin, sexual orientation, disability, or gender.

Not Hate Speech: Content that does not express hatred or target groups based on identity attributes. Note: Offensive language that contains profanity or insults but does not target identity groups is NOT considered hate speech.

Fox News Hate Speech.

Hateful: Language that explicitly or implicitly threatens or demeans a person or a group based upon a facet of their identity such as gender, ethnicity, or sexual orientation. This includes comments that express prejudice, discrimination, or hostility toward individuals or groups based on their protected characteristics.

Non-hateful: Comments that do not contain language that threatens or demeans individuals or groups based on their identity characteristics.

A.2. Medium Scope Definitions

These definitions include insults and threats regardless of identity targeting, but still require explicit negative content.

GameTox Multi-Category Definition.

Toxic: Any message that falls into one or more of these categories: (1) Hate and Harassment: Utterances with identity-based hate or harassment; (2) Threats: Utterances with threats of violence, physical harm, terrorism, or doxing; (3) Extremism: Utterances with extremist views or recruitment attempts; (4) Insults and Flaming: Insults or attacks on another player (not identity-based); (5) Other Offensive Texts: Any message that harms reasonable enjoyment of the game.

Not toxic: Utterances without any toxicity from the above categories.

OLID Offensive Language (Zampieri et al., 2019).

Offensive: This post contains offensive language or a targeted (veiled or direct) offense. This includes any form of non-acceptable language (profanity), insults, threats, and posts containing profane language or swear words.

Not Offensive: This post does not contain offense or profanity. The content may express opinions or criticism but does not include any form of offensive language, insults, or threats.

A.3. Broad Scope Definitions

These definitions capture any content that is “rude,” “disrespectful,” or “discouraging” - very broad criteria.

Gaming Toxicity (CONDA) (Weld et al., 2021).

Toxic: Utterances containing explicit or implicit toxicity in gaming context. *Explicit Toxicity:* Direct toxic expressions such as profanity, personal attacks, or insults directed at other players. *Implicit Toxicity:* Subtler forms including sarcasm, veiled insults, or coded language that requires conversational context to identify as harmful.

Not Toxic: Utterances related to gameplay actions, neutral communication, or content that does not discourage player engagement. General expressions of frustration not directed at specific players (e.g., “damn!” or “ugh”) are not toxic.

General Toxicity (Perspective API/Jigsaw) (cjadams et al., 2018).

A rude, disrespectful, or unreasonable comment that is likely to make someone leave a discussion.

Table 8. Definition scope classification. **Narrow** definitions require identity-based targeting (e.g., race, religion, gender) to qualify as toxic, excluding general insults. **Medium** definitions include direct insults, threats, and profanity regardless of identity targeting. **Broad** definitions capture any content that is rude, disrespectful, or discouraging, including implicit and contextual toxicity. When applied to general toxicity tasks, narrow definitions produce under-prediction while broad definitions produce over-prediction.

Definition	Scope
Twitter Hate Speech	Narrow
Fox News Hate Speech	Narrow
GameTox Toxicity	Medium
OLID Offensive	Medium
Gaming Toxicity (CONDA)	Broad
General Toxicity	Broad

A.4. Non-Safety Task Definitions

The following definitions were used in the generalization experiments reported in Appendix F. The irony and subjectivity definitions were used as the “aligned definition” prompt in the annotation experiments (Tables 24 and 25); the five stance definitions were used as the elicitation context for DSF on stance detection. All definitions operationalize the target concept following the framing of the original task papers and do not fit the narrow/medium/broad scope schema of Table 8, which applies only to the toxicity datasets.

Irony Detection (SemEval-2018 Task 3) (Van Hee et al., 2018).

Ironic: A statement where the intended meaning is opposite to or incongruent with the literal meaning, often used to express humor, criticism, or sarcasm. Irony involves a discrepancy between what is said and what is meant, or between expectations and reality.

Not Ironic: A statement where the intended meaning matches the literal meaning, with no incongruence between expression and intent.

Subjectivity Detection (Pang & Lee, 2004).

Subjective: A sentence that expresses personal opinions, evaluations, emotions, or speculations. Subjective sentences reflect the author’s point of view, feelings, or interpretations rather than objective facts.

Objective: A sentence that presents factual information that can be verified independently of the author’s personal opinion or emotional state.

Stance Detection (SemEval-2016 Task 6) (Mohammad et al., 2016). The stance task uses five targets, each with its own definition. Unlike the other tasks in our study, stance is framed as a choice between two explicit alternatives (*in favor / against*) rather than as classification against a single concept definition, as discussed in Appendix F.

Stance toward Legalization of Abortion. *In favor:* The tweet expresses support for or agreement with the legalization of abortion, including women’s right to choose. *Against:* The tweet expresses opposition to the legalization of abortion, including pro-life positions or moral objections.

Stance toward Atheism. *In favor:* The tweet expresses support for, agreement with, or positive views toward atheism or non-belief in God. *Against:* The tweet expresses opposition to, criticism of, or negative views toward atheism.

Stance toward Climate Change. *In favor:* The tweet expresses belief in or concern about climate change as a real and serious issue requiring action. *Against:* The tweet expresses skepticism about, denial of, or opposition to claims about climate change.

Stance toward Feminist Movement. *In favor:* The tweet expresses support for or agreement with the feminist movement and gender equality. *Against:* The tweet expresses opposition to, criticism of, or negative views toward the feminist movement.

Stance toward Hillary Clinton. *In favor:* The tweet expresses support for, agreement with, or positive views toward Hillary Clinton. *Against:* The tweet expresses opposition to, criticism of, or negative views toward Hillary Clinton.

B. Prompts

All prompts follow a consistent template structure. The base classification prompt presents the task, followed by the text to classify, and ends with an “Answer:” suffix. The confidence elicitation suffix (Figure 2) is appended to all classification prompts.

Experimental Conditions. Table 9 summarizes the 10 manually-specified prompting conditions together with the 2 DSPy automated optimization conditions used in our experiments.

Table 9. Description of experimental conditions.

Condition	Description	Aligned
zero_shot	No definition provided	–
aligned_definition	Definition matches dataset	Yes
few_shot	4 examples, no definition	–
few_shot_aligned_def	4 examples + aligned definition	Yes
misaligned_gaming_toxicity	Gaming toxicity definition	No
misaligned_general_toxicity	Perspective API definition	No
misaligned_gametox_toxicity	GameTox definition	No
misaligned_foxnews_hate_speech	Fox News hate speech definition	No
misaligned_twitter_hate_speech	Twitter hate speech definition	No
misaligned_olid_offensive	OLID offensive definition	No
dspy_optimized	DSPy-selected few-shot examples	–
dspy_aligned	DSPy-selected few-shot examples + aligned definition	Yes

B.1. DSPy Automated Optimization

We additionally evaluate two automated prompt optimization conditions using DSPy (Khattab et al., 2023), which selects few-shot demonstrations from a labeled validation pool (of 100 examples) to maximize accuracy. **DSPy Optimized** uses a bare classification signature with no human-written definition (automated counterpart to `few_shot`). **DSPy Aligned** additionally includes the dataset’s aligned definition in the instruction field (automated counterpart to `few_shot_aligned_def`). Optimization is run independently per model–dataset pair.

B.2. Continuation Prompt

To measure text-level familiarity (ROUGE-L), we split each text instance at the 40% mark by word count. We prompt the model with the first 40% (prefix) using the following template:

Continue the following passage faithfully: [PREFIX]

The model generates a continuation using greedy decoding (temperature 0.0). We then compute the ROUGE-L F1 score between the generated continuation and the remaining 60% of the ground truth text (suffix).

Additional memorization metrics. To confirm that ROUGE-L’s limitations as a memorization proxy do not drive our findings, we compute two semantically rich alternatives over the same continuation/suffix pairs. Both use the `all-MiniLM-L6-v2` encoder—the same encoder as DSF—so any sign difference between memorization and DSF reflects signal type rather than encoder choice. **Embedding Similarity** is the cosine similarity between the sentence embeddings of the generated continuation and the ground-truth suffix. **BERTScore** (Zhang et al., 2020) computes token-level F1 via pairwise cosine similarities of contextual token embeddings of the two texts. Both metrics are averaged at the model–dataset level before correlation analysis.

C. Mixed Effects Regression Models

We fit four mixed effects logistic regression models to analyze annotation correctness and error correction. All models include message-level random intercepts to account for repeated measures across experimental conditions. Models were estimated using Bayesian variational inference (`BinomialBayesMixedGLM` in `statsmodels`).

Model 1: Domain-Alignment Effects **Research Question:** What factors predict correct annotation?

Formula: $\text{correct} \sim \text{C}(\text{model}) + \text{C}(\text{condition}) + \text{C}(\text{domain}) + \text{is_aligned_defn} + \text{C}(\text{domain}):\text{is_aligned_defn} + (1|\text{message})$

Reference categories: Llama-3.1-8B (model), Forum (domain), Misaligned definition (condition)

Table 10. Model 1: Full results for domain-alignment effects.

Predictor	Coefficient	OR [95% CI]
Intercept	1.549	4.71 [4.63, 4.78]
<i>Model Effects (vs. Llama-8B):</i>		
DeepSeek-V3	0.499	1.65 [1.58, 1.72]
Llama-3.1-70B	0.237	1.27 [1.22, 1.32]
Mistral-7B	0.384	1.47 [1.42, 1.52]
Mistral-Small-24B	0.326	1.39 [1.33, 1.44]
Mixtral-8x7B	0.456	1.58 [1.52, 1.64]
<i>Domain Effects (vs. Forum):</i>		
Gaming	2.628	13.85 [13.31, 14.41]
News	−0.048	0.95 [0.92, 0.99]
Social Media	1.695	5.45 [5.27, 5.63]
<i>Alignment Effects:</i>		
is_aligned_defn	0.073	1.08 [1.04, 1.11]
Gaming × Aligned	0.419	1.52 [1.40, 1.65]
News × Aligned	0.645	1.91 [1.80, 2.02]
Social Media × Aligned	0.333	1.40 [1.32, 1.47]
Random Effects SD	1.156	3.18 [3.11, 3.24]

Model 2: Model-Alignment Interaction **Research Question:** Do different models benefit differently from aligned definitions?

Formula: $\text{correct} \sim \text{C}(\text{model}) + \text{C}(\text{condition}) + \text{C}(\text{domain}) + \text{is_aligned_defn} + \text{C}(\text{model}):\text{is_aligned_defn} + (1|\text{message})$

Table 11. Model 2: Model $\times$ Alignment interactions.

Model	Alignment Interaction	OR [95% CI]
Llama-3.1-8B (ref)	0.195	1.22 [1.18, 1.25]
Llama-3.1-70B	0.437	1.55 [1.44, 1.67]
DeepSeek-V3	0.320	1.38 [1.28, 1.48]
Mistral-Small-24B	0.202	1.22 [1.14, 1.32]
Mixtral-8x7B	0.100	1.11 [1.03, 1.19]
Mistral-7B	0.078	1.08 [1.01, 1.16]

Llama-3.1-70B benefits most from alignment (OR=1.55), despite having the lowest rescue rate. This suggests larger models have stronger priors that are harder to override, but can better utilize guidance when provided.

Model 3: Zero-Shot Predictor **Research Question:** Does zero-shot correctness predict prompted performance?

Formula: $\text{correct} \sim C(\text{model}) + C(\text{condition}) + C(\text{domain}) + \text{zero_shot_correct} + (1|\text{message})$

Table 12. Model 3: Zero-shot as predictor of prompted correctness.

Predictor	Coefficient	OR [95% CI]
Zero-shot Correct	1.861	6.43 [6.28, 6.58]
<i>Domain Effects (vs. Forum):</i>		
Gaming	2.124	8.36 [8.01, 8.73]
News	0.177	1.19 [1.15, 1.24]
Social Media	1.487	4.43 [4.26, 4.60]

Model 4: Rescue Prediction **Research Question:** Among zero-shot errors, what predicts whether prompting will correct the error?

Formula: $\text{rescued} \sim C(\text{model}) + C(\text{condition}) + C(\text{domain}) + \text{zs_conf_std} + (1|\text{message})$

This model is fit only on cases where the model was wrong in zero-shot (N=47,672 errors).

Table 13. Model 4: Predicting error correction (rescue) among zero-shot errors.

Predictor	Coefficient	OR [95% CI]
ZS Confidence (standardized)	-0.171	0.84 [0.82, 0.87]
<i>Model Effects (vs. Llama-8B):</i>		
DeepSeek-V3	0.634	1.89 [1.75, 2.03]
Llama-3.1-70B	0.111	1.12 [1.04, 1.20]
Mistral-7B	-0.071	0.93 [0.87, 1.00]
Mistral-Small-24B	0.514	1.67 [1.57, 1.79]
Mixtral-8x7B	0.296	1.35 [1.25, 1.45]
<i>Domain Effects (vs. Forum):</i>		
Gaming	0.982	2.67 [2.47, 2.89]
News	0.580	1.79 [1.68, 1.89]
Social Media	1.331	3.78 [3.56, 4.02]

D. Additional Experimental Results

Table 14. Detailed breakdown of model accuracy (%) under specific misaligned definitions. These values are averaged into the single “Misaligned” column in Table 3. Columns are ordered by definition scope (Narrow → Broad). Models below the horizontal rule are not included in regression analyses; Cond. AUC is for the six open-weights models.

Model	Misaligned Definitions					
	Twitter Hate	FoxNews Hate	OLID Off.	GameTox Tox.	General Tox.	Gaming Tox.
DeepSeek-V3	79.2	82.0	82.6	80.8	82.0	77.4
Mixtral-8x7B	79.0	83.3	82.1	80.9	83.3	78.7
Mistral-7B	79.5	83.8	84.0	80.3	84.5	76.5
Mistral-Small-24B	77.2	83.1	82.9	80.3	83.5	77.3
Llama-3.1-70B	77.5	81.4	80.0	78.3	78.1	73.0
Llama-3.1-8B	77.3	79.8	78.2	76.8	76.9	72.7
GPT-4o-mini	79.8	82.3	82.1	82.0	81.6	78.5
Llama-3.3-70B	78.6	83.0	81.5	77.8	78.9	74.4
Qwen-2.5-72B	78.6	84.3	83.9	79.7	81.0	79.3
Cond. Avg.	78.5	82.6	81.9	79.7	81.1	76.4
Cond. AUC	.790	.850	.867	.817	.857	.841

Table 15. Accuracy (%) by Dataset and Misaligned Condition. Bold = best for that dataset. “N/A” indicates the aligned (non-misaligned) condition.

Dataset	foxnews	gametox	gaming	general	olid	twitter
FoxNews	N/A	75.6	61.6	70.6	73.8	78.4
GameTox	87.2	N/A	90.6	90.6	90.9	85.0
Jigsaw	75.5	75.8	70.4	N/A	77.4	73.0
OLID	75.8	79.7	79.7	79.9	N/A	74.3
Twitter	88.7	89.1	80.8	85.7	84.4	N/A

Table 16. Model sensitivity: best accuracy (across all conditions) minus worst accuracy under misaligned definitions. Models below the horizontal rule are not included in regression analyses.

Model	Best Acc.	Worst Acc.	Sensitivity
Llama-3.1-70B	82.1	73.0	9.1 (highest)
Mistral-7B	84.5	76.5	8.0
Llama-3.1-8B	80.1	72.7	7.4
DeepSeek-V3	83.8	77.4	6.4
Mistral-Small-24B	83.5	77.2	6.3
Mixtral-8x7B	83.3	78.7	4.6 (lowest)
Llama-3.3-70B	83.0	74.4	8.6
GPT-4o-mini	84.1	78.5	5.6
Qwen-2.5-72B	84.3	78.6	5.7

Table 17. Corruption rate (%) by model and misaligned condition. High corruption indicates that correct predictions are frequently flipped to incorrect ones when the definition changes. Models below the horizontal rule are not included in regression analyses.

Model	Twitter	FoxNews	OLID	GameTox	General	Gaming
DeepSeek-V3	14.7	10.7	6.4	7.5	7.4	12.1
Llama-3.1-70B	16.8	12.3	6.0	7.5	8.3	13.3
Llama-3.1-8B	14.0	10.9	11.1	10.4	7.6	12.4
Mistral-7B	8.4	4.6	4.0	8.2	2.2	9.9
Mistral-Small-24B	21.0	11.3	5.0	8.2	3.1	10.0
Mixtral-8x7B	9.0	6.5	6.7	5.6	4.8	10.4
GPT-4o-mini	13.2	10.3	5.7	10.9	5.7	9.0
Llama-3.3-70B	15.2	10.0	5.4	6.9	8.3	12.0
Qwen-2.5-72B	12.6	7.2	4.6	7.4	8.6	9.6

Table 18. Rescue rate by model on the Jigsaw Unintended Bias dataset (Borkan et al., 2019), which was designed to reduce identity-group annotation bias. Rescue rate is $P(\text{correct} \mid \text{prompted, zero-shot wrong})$, computed over all non-zero-shot conditions (aligned definition, few-shot, few-shot+definition, DSPy Optimized, DSPy Aligned, and five misaligned conditions), matching the methodology of Table 6. All models remain well below 50%, confirming that decision stickiness is not an artifact of Jigsaw’s known label-bias limitations.

Model	Rescue Rate	95% CI	N (ZS err. obs.)
DeepSeek-V3	38.2%	[36.4, 40.1]	2,640
Mistral-Small-24B	35.6%	[33.9, 37.4]	2,810
Llama-3.1-8B	34.3%	[32.5, 36.0]	2,790
GPT-4o-mini	31.4%	[29.6, 33.3]	2,350
Mixtral-8x7B	31.0%	[29.3, 32.8]	2,790
Llama-3.1-70B	30.8%	[29.1, 32.5]	2,870
Llama-3.3-70B	28.2%	[26.6, 29.9]	2,790
Mistral-7B	27.8%	[26.1, 29.5]	2,680
Qwen-2.5-72B	24.9%	[23.2, 26.7]	2,330

D.1. Multi-Turn Rescue Experiment

A natural follow-up to the one-shot rescue analysis (Sec. 4.2) is whether iterative, history-aware prompting can break decision stickiness. For each model–dataset pair we sampled 100 zero-shot errors and applied a 3-step rescue sequence: Turn 1 adds a few labeled examples, Turn 2 adds the aligned task definition, and Turn 3 asks the model to reconsider. At each turn the model is shown its previous answers and confidence scores, so later prompts see the full history.

Table 19 reports cumulative rescue rates by model, both overall and restricted to high-confidence errors. Iterative prompting helps modestly — particularly when the aligned definition is added at Turn 2 — but rescue plateaus well below the one-shot rate of 34.8% reported in Table 6, and high-confidence errors remain almost entirely uncorrected (8.5% rescued after three turns).

Table 19. Cumulative rescue rates over three multi-turn correction attempts. “High-conf” columns restrict to zero-shot errors with confidence > 0.9 .

Model	Rescue@1	Rescue@2	Rescue@3	High-conf.@1	High-conf.@2	High-conf.@3
DeepSeek-V3	8.2	20.8	27.4	0.0	5.1	6.8
Llama-3.1-70B	10.6	14.6	19.0	2.9	5.0	8.0
Llama-3.1-8B [†]	21.7	35.8	35.8	0.0	16.7	16.7
Mistral-7B	3.0	12.3	12.7	1.2	8.5	9.1
Mistral-Small-24B	7.0	12.4	12.9	4.1	9.6	9.6
Mixtral-8x7B	5.8	17.1	18.5	1.9	6.5	7.4
Pooled	7.5	16.3	18.7	2.0	7.1	8.5

[†]78.8% of sampled items dropped due to classification refusals.

E. Familiarity Analysis Details

E.1. DSF Robustness Across Embedding Models

The DSF score is a cosine similarity in an embedding space, so its value depends on the encoder used. To rule out that our results are an artifact of choosing any single embedding model, we recompute DSF using six diverse sentence-embedding models spanning small (MiniLM, mpnet) to large (bge-large, e5-large, instructor-large) open-source encoders, plus a proprietary OpenAI model (text-embedding-3-small). The full model identifiers are:

- MiniLM: sentence-transformers/all-MiniLM-L6-v2
- MPNet: sentence-transformers/all-mpnet-base-v2
- BGE-large: BAAI/bge-large-en-v1.5
- E5-large: intfloat/e5-large-v2
- Instructor-large: hkunlp/instructor-large
- OpenAI: text-embedding-3-small

We then define *consensus DSF* for each model–dataset cell as the unweighted mean DSF across the six encoders, and adopt it as the primary DSF metric in the main text.

The DSF analysis covers all nine evaluated models across six datasets (the five primary datasets plus Jigsaw Unintended Bias (Borkan et al., 2019)), yielding $N = 54$ model–dataset pairs. Model performance is reported in Table 3.

Per-embedding partial correlations. Table 20 reports the partial correlation between DSF and zero-shot accuracy (controlling for dataset) separately for each embedding choice. All six embeddings yield a positive partial correlation, ranging from +0.30 to +0.49. Larger encoders (bge-large, e5-large, OpenAI) tend to yield slightly stronger correlations than the compact MiniLM. The consensus row (+0.41) sits at the upper end of this range, indicating that averaging across encoders denoises rather than dilutes the signal.

Table 20. DSF–accuracy correlations across six sentence-embedding models on the extended $N = 54$ panel (zero-shot accuracy, partial correlation controls for dataset). Consensus DSF is the per-cell mean across embeddings, and is the metric reported in the main text.

Embedding model	Raw r	Partial r
all-MiniLM-L6-v2	+0.73	+0.36
all-mpnet-base-v2	+0.77	+0.33
BAAI/bge-large-en-v1.5	+0.74	+0.49
intfloat/e5-large-v2	+0.60	+0.43
hkunlp/instructor-large	+0.72	+0.30
OpenAI text-embedding-3-small	+0.72	+0.43
Consensus (mean across 6)	+0.74	+0.41

Pairwise embedding agreement. DSF values are highly concordant across embedding choices. Table 21 reports the Pearson correlation between DSF score vectors produced by different embedding models on the $N = 54$ panel; all pairwise correlations exceed 0.87, and most exceed 0.95. This indicates that the variation in Table 20 reflects what each embedding emphasizes (e.g., lexical vs. semantic, general vs. domain-tuned) rather than disagreement about which model–dataset pairs are high- vs. low-DSF.

Table 21. Pairwise Pearson correlations between DSF score vectors across six embedding models, on the $N = 54$ panel. All embeddings agree closely on the relative ordering of model–dataset cells.

	MiniLM	mpnet	bge	e5	instr.	OpenAI
MiniLM	1.00	0.97	0.98	0.93	0.97	0.97
mpnet		1.00	0.96	0.88	0.97	0.97
bge-large			1.00	0.95	0.98	0.99
e5-large				1.00	0.94	0.93
instructor					1.00	0.98
OpenAI						1.00

Takeaway. The DSF–accuracy association is not an artifact of a particular encoder. Across six embedding models the partial correlation remains positive and meaningful, and the embeddings closely agree on which model–dataset pairs score high or low. We therefore report consensus DSF in the main text as a stable, encoder-agnostic operationalization of the metric.

E.2. Min-K% Prob as a Memorization Baseline

Min-K% Prob (Shi et al., 2024) computes the average log probability of the $k\%$ least likely tokens in a text. Lower (more negative) values indicate the model finds certain tokens surprising, suggesting the text was *not* memorized during training. Computing Min-K% requires per-token log probabilities, which are unavailable for most frontier API models in our evaluation. We therefore use ROUGE-L as the primary memorization baseline, as it is computable via text generation for any model.

To verify that this choice does not change our qualitative conclusions, we applied Min-K% ($k = 20\%$) to the two open-weights models in our panel for which log probabilities are available (Llama-3.1-8B and Mistral-7B, base versions) across the five paper datasets ($N = 10$ model–dataset pairs). Table 22 reports mixed-effects model coefficients (Accuracy $\sim$ metric + (1 | dataset) + (1 | model)) for ROUGE-L, Min-K%, and DSF on this subset. Min-K% and ROUGE-L are directionally consistent—both show negative standardized coefficients while DSF remains positive—confirming that our conclusions do not depend on the choice of memorization metric.

Table 22. Mixed-effects model coefficients for memorization metrics and DSF on the $N = 10$ subset (Llama-3.1-8B and Mistral-7B $\times$ 5 paper datasets) where Min-K% Prob is computable. Standardized coefficients allow cross-metric comparison of effect size.

Metric	Std. Coef.	Raw Coef.	p
Text Memorization (ROUGE-L)	−0.755	−2.479	0.001
Memorization (Min-K% Prob)	−0.596	−0.044	0.036
Definition Familiarity (DSF)	+0.838	+0.425	< 0.001

E.3. Semantic Memorization Metrics: BERTScore and Embedding Similarity

ROUGE-L captures lexical overlap via the longest common subsequence and may therefore underestimate memorization when a model reproduces the *content* of a passage with paraphrased surface form. We additionally evaluate two semantically richer alternatives that operate on the same generated continuations as ROUGE-L, so the comparison isolates the scoring function rather than the generation procedure.

BERTScore. BERTScore (Zhang et al., 2020) replaces n -gram matching with contextual token-embedding similarity. Each token in the generated continuation $\hat{y}$ and the reference suffix y is encoded with a pretrained contextual encoder, and tokens are greedily aligned by maximum cosine similarity:

$$R_{\text{BERT}} = \frac{1}{|y|} \sum \max_{\hat{y}_j \in \hat{y}} \mathbf{y}_i^\top \hat{\mathbf{y}}_j, \quad P_{\text{BERT}} = \frac{1}{|\hat{y}|} \sum \max_{y_i \in y} \mathbf{y}_i^\top \hat{\mathbf{y}}_j,$$

with $F_{1,\text{BERT}}$ as their harmonic mean. Because the embeddings are contextual, BERTScore credits paraphrases and near-synonyms that ROUGE-L would penalize. We use `microsoft/deberta-xlarge-mnli` as the encoder (the default recommended by Zhang et al. (2020) for English) and report $F_{1,\text{BERT}}$ as the primary score.

Embedding cosine similarity. Embedding similarity moves one level higher, comparing whole-sentence representations rather than token alignments:

$$\text{EmbSim}(\hat{y}, y) = \frac{\phi(\hat{y})^\top \phi(y)}{\|\phi(\hat{y})\| \|\phi(y)\|},$$

where $\phi(\cdot)$ is a sentence encoder. This is the loosest of the three notions of textual overlap: a continuation that captures the gist of the reference but rewords it freely can still score high. To ensure that any difference between EmbSim and DSF reflects *what* is being measured (continuation–reference alignment vs. definition–self-description alignment) rather than *which encoder* does the measuring, we use the same `all-MiniLM-L6-v2` encoder as in DSF.

Spectrum of memorization sensitivity. Taken together, ROUGE-L, BERTScore, and EmbSim form a progression from purely lexical to fully semantic notions of reproduction. If text memorization were driving annotation accuracy and ROUGE-L were simply too crude to detect it, we would expect the correlation with accuracy to strengthen (or at least become positive) as we relax the lexical constraint. Table 4 shows the opposite: all three text-memorization metrics yield negative partial correlations with accuracy after controlling for dataset, while DSF remains positive. The negative direction is preserved across the full lexical–semantic spectrum, indicating that the issue is not ROUGE-L’s lexical sensitivity but that text reproduction itself, however measured, does not predict annotation performance.

E.4. Definition-Specific Familiarity (DSF) Variants

In the main text, we define **Definition-Specific Familiarity (DSF)** as the cosine similarity between the embedding of the dataset’s ground-truth definition and the model’s own description of the concept. Figure 6 illustrates this primary metric; below, we detail it alongside two alternative variants we explored.

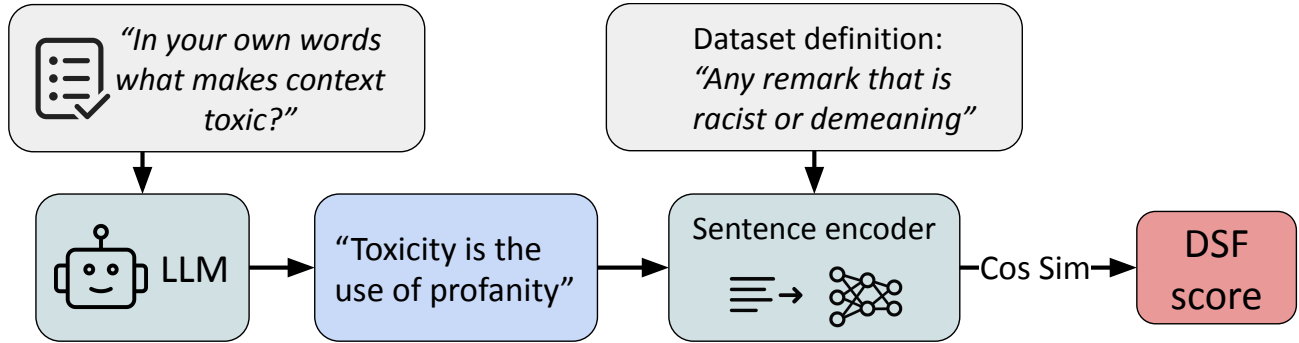


Figure 6. **Conceptual overview of Definition-Specific Familiarity (DSF).** We first ask the model to explain the target concept in its own words. We treat the model-generated explanation as an operational proxy for its internal concept. We then encode the explanation and the dataset’s reference definition with the same sentence encoder and compute their cosine similarity. A higher DSF score indicates closer semantic alignment between the model’s elicited concept and the definition expected by the dataset. A lower score indicates poorer alignment.

DSF (Primary Metric). This is the metric reported in the main results. We prompt the model with a neutral instruction to describe the concept: “*You are documenting your internal understanding of a labeling concept. Concept: [Concept Name]. Describe, in your own words, what content qualifies as this concept.*” We then compute the cosine similarity between the embedding of this generated response and the embedding of the dataset’s full definition. We found this approach correlated most strongly with model performance.

Alternative Variant: Positive DSF. This variant isolates the “positive” class definition (e.g., what *is* toxicity) from the “negative” class. It uses **multi-prompt elicitation** to reduce variance, averaging similarities across three prompt styles: *Direct* (“You are documenting...”), *Expert* (“As an expert annotator...”), and *Examples* (“Without examples, describe...”).

Alternative Variant: Enhanced DSF. This metric extends Positive DSF by adding **domain contextualization** (e.g., “*in online gaming chat*” for GameTox) and **negative class alignment** (comparing the model’s description of what the concept is *not* to the dataset’s negative definition). The final score is the average of the positive and negative alignment scores.

Table 23. Familiarity metrics by dataset (averaged across the six open-weights models, zero-shot). DSF is reported as consensus DSF across six sentence encoders. Jigsaw has the lowest DSF and is the hardest dataset; GameTox has the highest accuracy with negligible text memorization, illustrating that conceptual alignment, not memorization, tracks performance.

Dataset	Domain	ROUGE-L	Consensus DSF	Accuracy
Twitter	Social Media	0.045	0.804	0.901
GameTox	Gaming	0.001	0.784	0.902
OLID	Social Media	0.063	0.710	0.751
FoxNews	News	0.061	0.652	0.723
Jigsaw	Forum	0.084	0.458	0.705

F. Generalization Beyond Toxicity

To probe whether our two main findings—the positive DSF–accuracy association (RQ1) and decision stickiness (RQ2)—extend beyond safety tasks, we conduct supplementary experiments on two binary classification tasks: *irony detection* (SemEval-2018 Task 3; Van Hee et al. 2018) and *subjectivity detection* (Pang & Lee, 2004). Both use explicit human-authored definitions and binary labels, making them structurally comparable to the toxicity setup. We evaluate the same 9 models from Table 2 on 1,000 instances per dataset–model pair. Because this extension contains only two datasets and fewer prompting conditions, we treat it as a generalization check rather than evidence of universality.

DSF extends directionally beyond toxicity. Across the 18 model–dataset pairs, DSF is positively associated with zero-shot accuracy after controlling for dataset (partial $r = +0.343$). This matches the direction of the main result, suggesting that DSF captures model–task fit beyond toxicity, although the two-dataset sample makes this evidence necessarily preliminary.

Decision stickiness persists beyond toxicity. Few-shot prompting rescues 45.0% (95% CI [43.2%, 46.8%]) of 3,052 zero-shot errors, compared with 34.8% on the toxicity tasks (Table 6). Thus, even where examples help more, most initial errors remain uncorrected. Tables 24 and 25 report condition-level accuracy and per-model rescue rates.

Table 24. Accuracy (%) by condition on the two non-safety datasets (irony and subjectivity), averaged across the two datasets. Sampling and inference settings match the main experiments.

Model	Zero-Shot	Aligned Def	Few-Shot
Llama-3.3-70B	73.3	70.4	86.2
Mixtral-8x7B	71.2	76.1	84.2
Llama-3.1-70B	71.0	68.9	85.2
Mistral-7B	68.8	65.9	73.9
DeepSeek-V3	68.1	72.4	76.9
Mistral-Small-24B	63.2	71.8	79.3
GPT-4o-mini	62.7	73.7	81.9
Qwen-2.5-72B	57.1	60.7	69.7
Llama-3.1-8B	50.2	57.1	72.1
Cond. Avg	65.1	68.5	78.8

Table 25. Rescue rate by model in the few-shot condition on the two non-safety datasets (irony and subjectivity). Rescue rate is $P(\text{correct} \mid \text{few-shot, zero-shot wrong})$, computed pooled across the two datasets.

Model	Rescue Rate	95% CI	N (ZS err.)
Mixtral-8x7B	59.9%	[53.9, 65.7]	277
Llama-3.1-8B	52.8%	[48.3, 57.3]	498
Llama-3.1-70B	52.8%	[46.8, 58.6]	290
GPT-4o-mini	52.3%	[47.1, 57.4]	373
Llama-3.3-70B	50.6%	[44.4, 56.7]	267
Mistral-Small-24B	47.1%	[41.9, 52.4]	367
DeepSeek-V3	37.6%	[32.3, 43.2]	319
Qwen-2.5-72B	27.9%	[23.6, 32.6]	401
Mistral-7B	21.5%	[16.7, 27.0]	260
Overall	45.0%	[43.2, 46.8]	3,052

Stance detection: a structurally different task. As a further check, we ran a preliminary experiment on stance detection (SemEval-2016 Task 6; Mohammad et al. 2016) across 5 targets and 8 models. The DSF direction remains positive (partial $r = +0.295$, $N = 40$ target–model pairs). Stance detection differs structurally from our other tasks, however: it requires choosing between two explicit alternatives (“in favor” / “against”) rather than classifying content against a single concept definition, which may introduce additional sources of variation beyond conceptual alignment around the definition itself. We therefore treat stance as directionally consistent but not as clean a test of the DSF–accuracy relationship as irony and subjectivity, and we report it here only as supporting evidence.

Takeaway. Irony and subjectivity show the same qualitative pattern as the primary experiments: better definition alignment accompanies higher accuracy, while most initial errors resist correction. These results broaden the evidence beyond toxicity, but the larger and more controlled toxicity experiments remain the basis of our main claims.

G. Model and Dataset Use Across Experiments

Our experiments use a two-tier structure for both models and datasets, summarized in Table 26.

The main-paper descriptive RQ2/RQ3 result tables and figures (Tables 6, 7 and Figs. 4, 5) report values aggregated across all nine evaluated models (Llama-3.1-8B, Llama-3.1-70B, Llama-3.3-70B, Mistral-7B, Mistral-Small-24B, Mixtral-8x7B, DeepSeek-V3, GPT-4o-mini, Qwen-2.5-72B). All mixed-effects regression analyses (Table 5 in the main paper and Tables 10–13 in the appendix) and the multi-turn rescue analysis (Table 19) are scoped to the original six open-weights models (Llama-3.1-8B, Llama-3.1-70B, Mistral-7B, Mistral-Small-24B, Mixtral-8x7B, DeepSeek-V3) to keep statistical inference tied to the original six-model design; GPT-4o-mini, Llama-3.3-70B, and Qwen-2.5-72B contribute to the descriptive and performance tables but are excluded from the mixed-effects regressions.

The dataset side follows the same pattern. The five primary toxicity datasets (Twitter Hate, OLID, GameTox, Fox News, Jigsaw Toxic Comments) are the analytic core. The three supplementary datasets—Jigsaw Unintended Bias, SemEval Irony, and Subjectivity—serve targeted checks: Jigsaw Unintended Bias tests robustness to label-bias concerns, while irony and subjectivity probe whether the RQ1 and RQ2 patterns extend beyond toxicity (Appendix F).

Table 26. Coverage by analysis. “6 core” and “9 models” denote the model tiers defined above; “5 main” denotes the primary toxicity datasets; Jigsaw UB = Jigsaw Unintended Bias.

Coverage	Where it applies
9 models $\times$ 5 main datasets	Tables 6, 7; Figs. 4, 5; Table 3; Tables 14, 16, 17 (extended models shown below midrule)
6 core models $\times$ 5 main datasets	Tables 5, 10–13, 15, 19, 23
9 models $\times$ 5 main + Jigsaw UB ($N=54$)	Tables 4, 20, 21 (familiarity / DSF)
9 models $\times$ Jigsaw UB only	Table 18 (rescue-rate robustness check)
9 models $\times$ Irony + Subjectivity	Tables 24, 25 (non-safety generalization)
Llama-3.1-8B + Mistral-7B (base) $\times$ 5 main	Table 22 (Min-K%, requires logprob access)